AI-Assisted Web Security Audit Report

Generated: 2026-11-06 05:26:30
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1. Port and Service Audit

Target: 127.0.0.1 (resolved: 127.0.0.1)

Scan time: 2026-11-06T05:24:42

Environment context: Loopback scan inside the hosted notebook runtime. Any identified listener may be part of Google Colab infrastructure; this scan does not establish exposure of the user's own computer or production application.

Port	Protocol	State	Service
8080	TCP	open	http-proxy

Port 8080 listener attribution: /tools/node/bin/node /datalab/web/app.js. Context: Likely Google Colab-managed web infrastructure; not identified as the project application.

Interpretation: an open port is an observed listening service, not by itself a confirmed vulnerability. Confirm service ownership, intended purpose, exposure scope, authentication, patching, and access controls before assigning severity.

Recommendation: verify business need for each service, restrict access with firewall/security-group rules, and keep required services patched. For a hosted notebook loopback scan, do not attribute platform-managed services to the user's own application.

2. HTTP Rate-Limit Indicator Check

Request	HTTP Status	Observed indicators
1	404	Not observed
2	404	Not observed
3	404	Not observed
4	404	Not observed
5	404	Not observed

Interpretation: Assessment inconclusive: one or more requests returned HTTP 404, so the tested path was not found. No conclusion about application rate limiting can be drawn from this path. Repeat only against a documented, known-valid endpoint.

3. AI Traffic Analysis

Predicted class	Flow count
BENIGN	2003
DoS	2000
PortScan	1997

Total analyzed flows: 6000. Predictions are model-based and should be reviewed alongside network and host context.

Model holdout accuracy: 99.38%. This score was measured on the stratified held-out test split during model evaluation.

Accuracy on uploaded labelled flows: 99.38% (5,963 correct out of 6,000 labelled flows).

4. General Mitigation Recommendations

- Close unnecessary services and expose only required ports.
- Restrict administrative and sensitive services to trusted networks or VPN access.
- Apply rate limits at the application, API gateway, reverse proxy, or web-server layer.
- Use firewall rules, monitoring, patching, and alerting as layered controls.
- Investigate ML alerts with logs and packet/flow context before taking action.

Scope note: This report reflects only the checks and uploaded flow records processed in this dashboard session. A localhost scan in Google Colab examines the hosted notebook runtime, not the user's personal computer or automatically the user's deployed application. Process attribution is best-effort. An open port is not by itself a vulnerability, and absence of rate-limit headers is not proof that rate limiting is absent. This report is not a certification or proof that a target is secure.